



NORTHERN UNIVERSITY

OF BUSINESS & TECHNOLOGY KHULNA

Software development 3 : System Design

Course Code: CSE 4104

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Project Overview

The SaaS in Business Management System is an AI-powered WordPress plugin designed to help small and medium-sized businesses manage their daily operations through a centralized platform. The system provides tools for inventory management, sales tracking, expense monitoring, and business performance analysis.

By integrating Artificial Intelligence, the system can analyze business data and generate insights, recommendations, and reports to support informed decision-making. The project aims to reduce manual workload, improve operational efficiency, and assist business owners in managing their businesses more effectively.

Key Features

- AI-powered business analytics dashboard
- Inventory and stock management
- Sales tracking and reporting
- Expense management

- AI-generated business insights and recommendations
- Business performance monitoring
- Automated reporting system

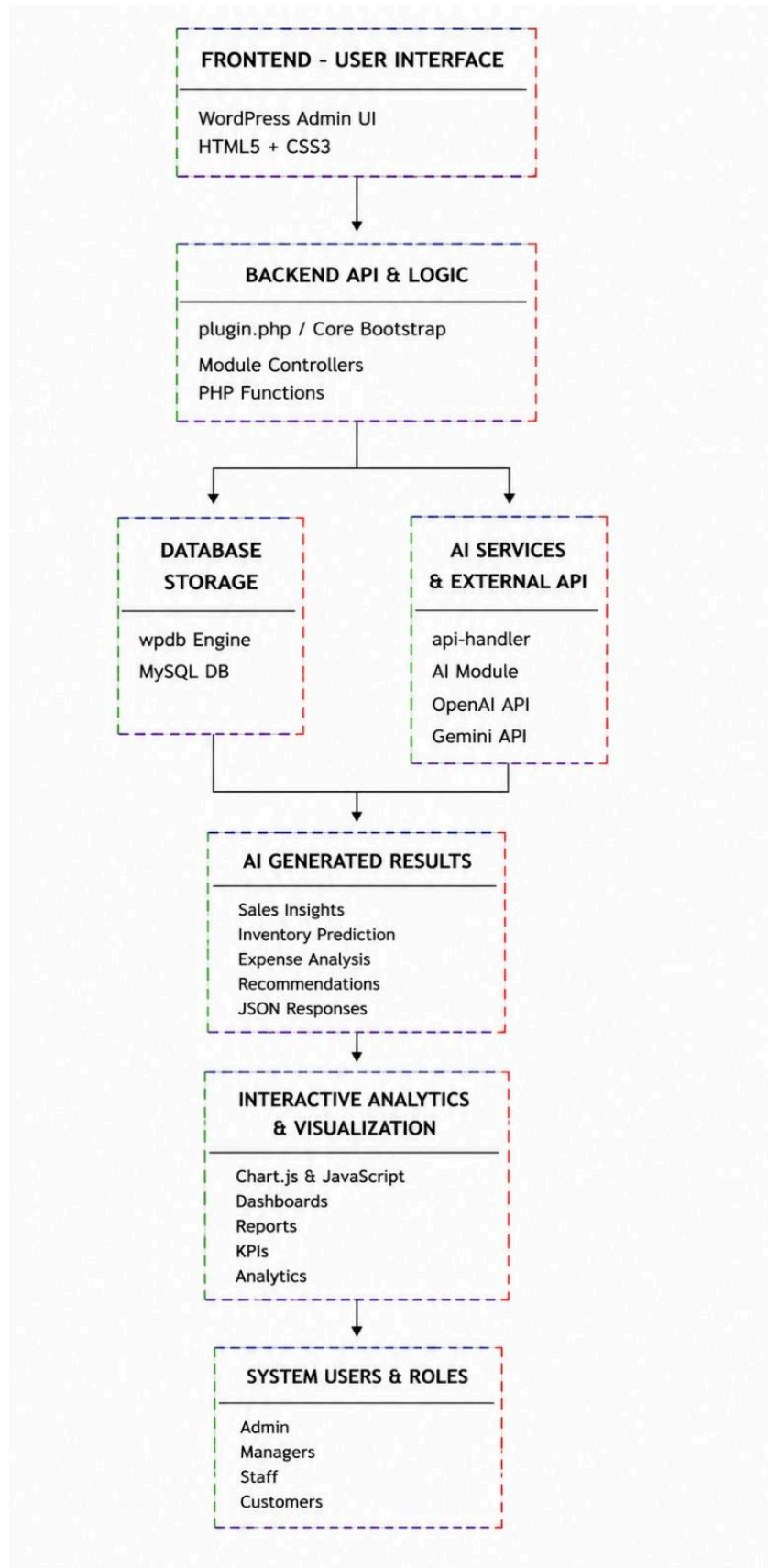
Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP (WordPress Plugin Development)
- **Database:** MySQL
- **AI Integration:** OpenAI API / Google Gemini API
- **Platform:** WordPress

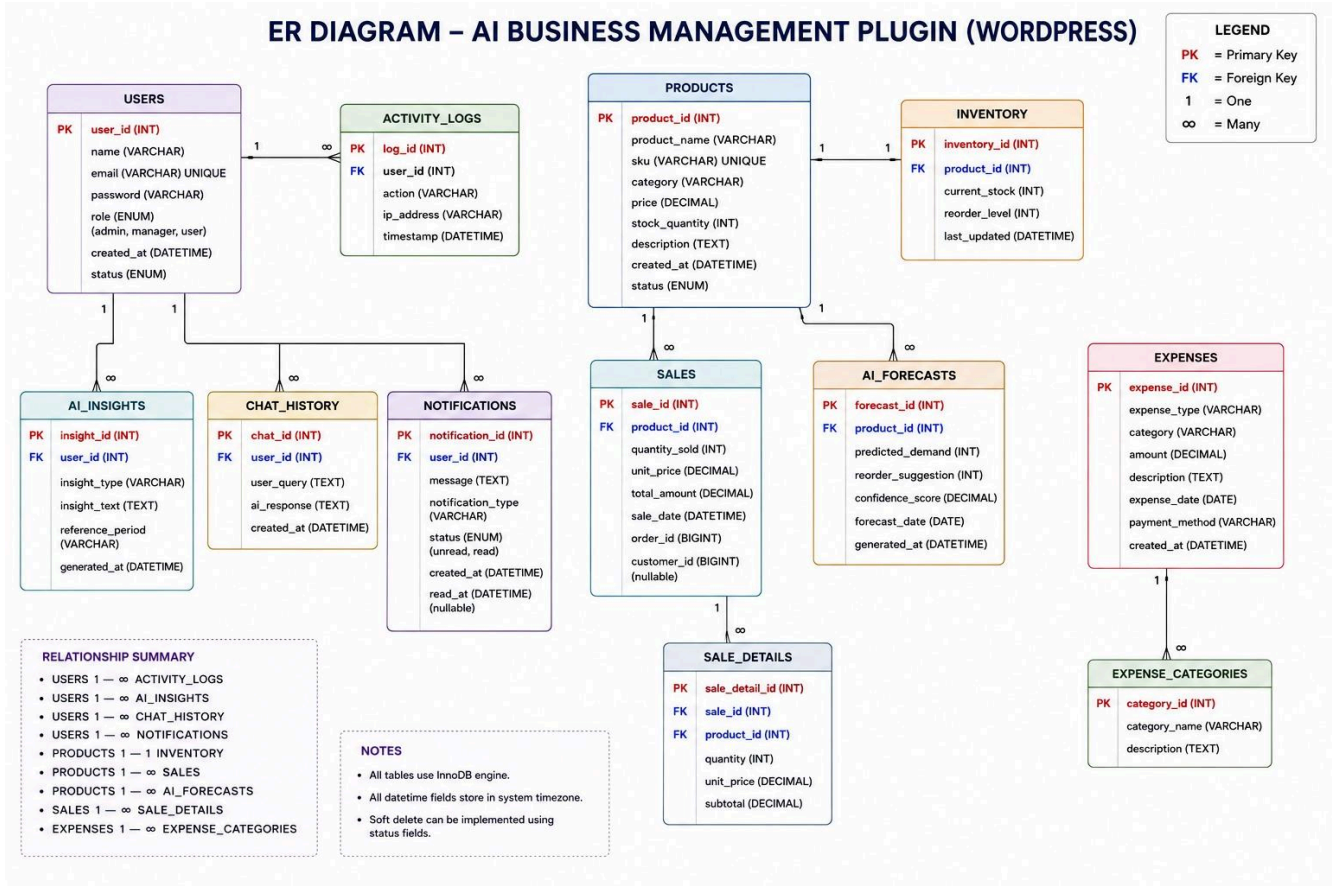
Expected Outcome

The system will provide business owners with a smart and user-friendly management solution that combines business operations with AI-powered analytics. It is expected to improve productivity, reduce errors, streamline business processes, and support data-driven decision-making.

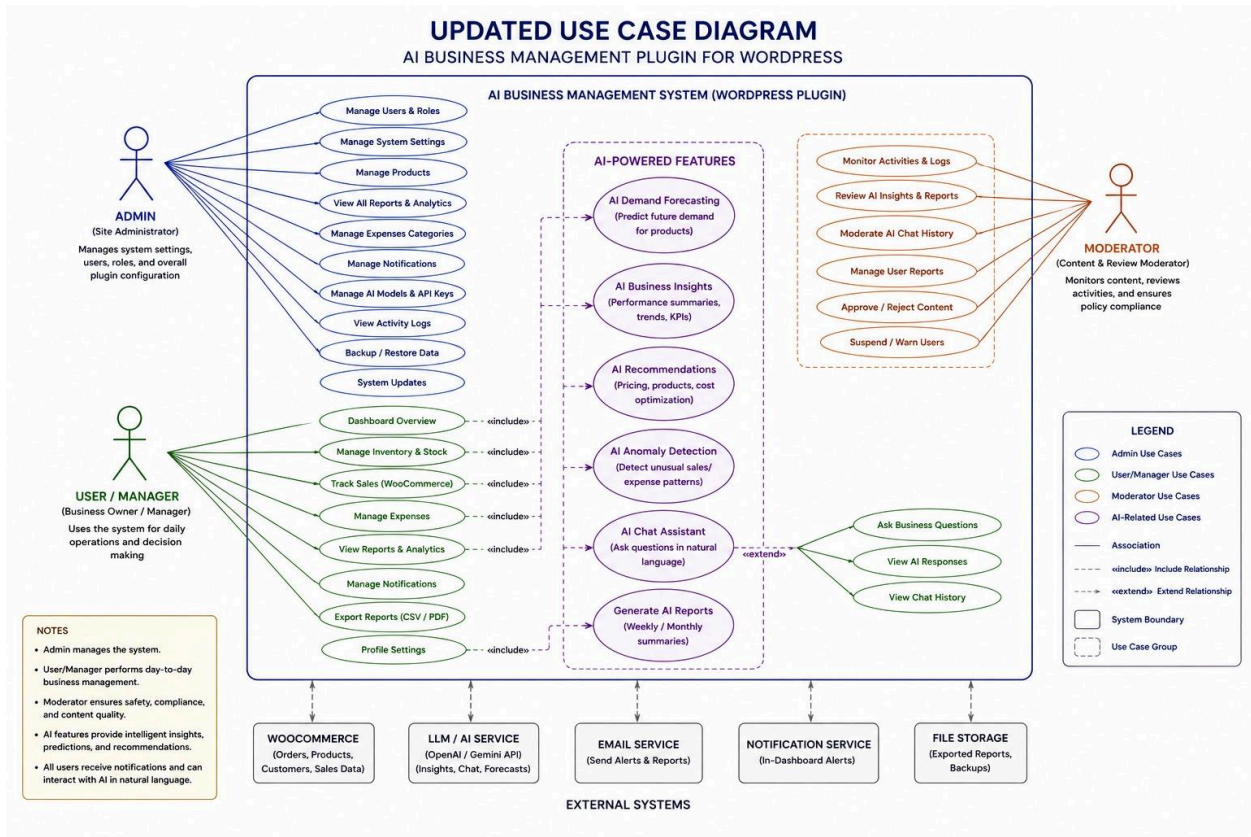
System Architecture Diagram



ER Diagram

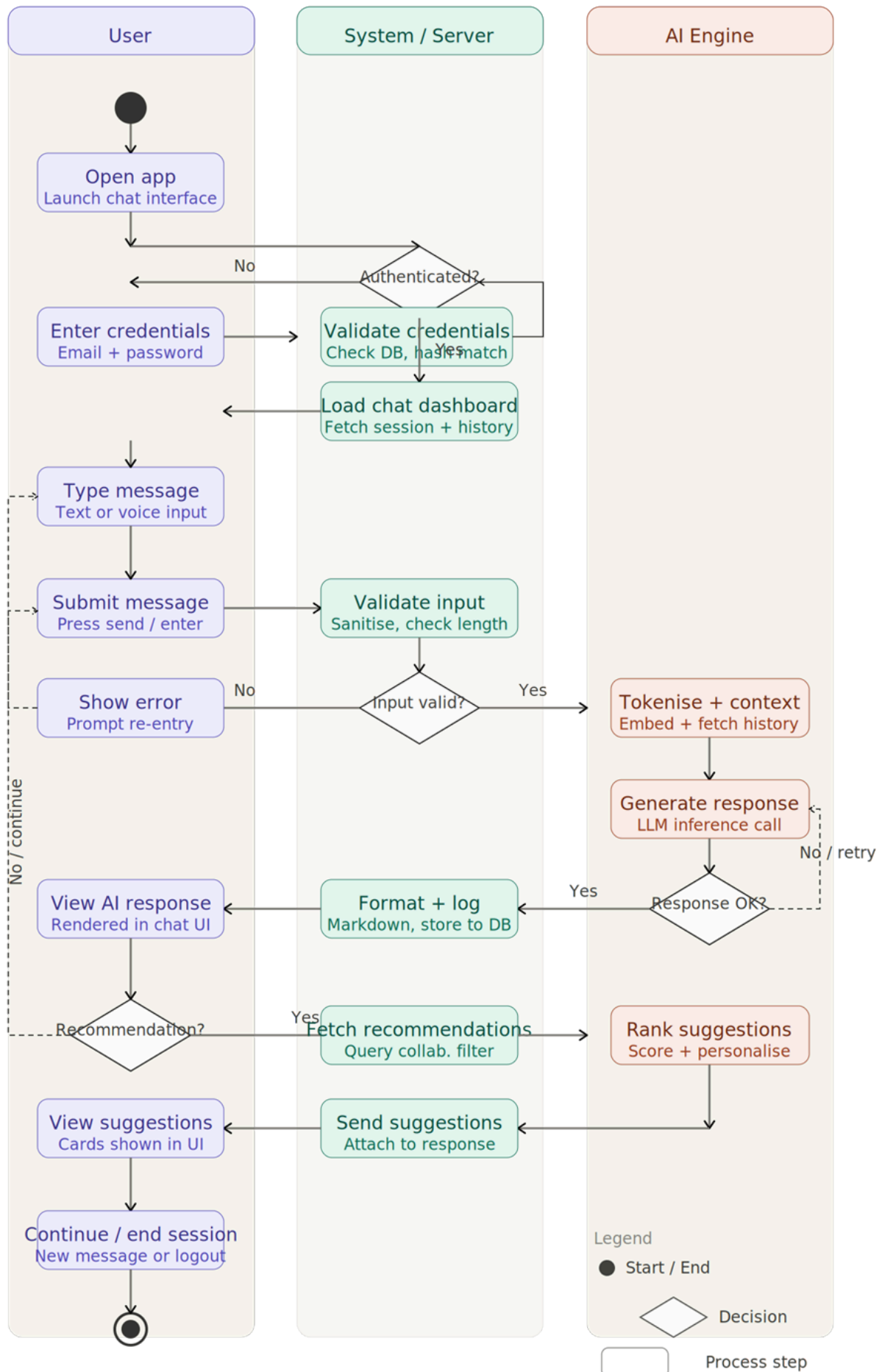


Updated Use Case Diagram



Activity Diagram

AI Chat Workflow



API Design Document

Authentication APIs

POST /api/auth/login

Purpose: Authenticate admin user and provide access to the system.

Input:

```
{
  "email": "admin@example.com",
  "password": "password123"
}
```

Expected Output:

```
{
  "status": "success",
  "message": "Login successful"
}
```

POST /api/auth/logout

Purpose: End the current user session.

Input:

```
{
  "user_id": 1
}
```

Expected Output:

```
{
  "status": "success",
  "message": "Logged out successfully"
}
```

Inventory Management APIs

GET /api/inventory

Purpose: Retrieve all inventory items.

Input: None

Expected Output:

```
{
  "products": [
    {
      "id": 1,
      "name": "Product A",
      "stock": 120
    }
  ]
}
```

POST /api/inventory/add

Purpose: Add a new inventory item.

Input:

```
{
```

```
"product_name": "Product A",  
"quantity": 100  
}
```

Expected Output:

```
{  
  "status": "success",  
  "message": "Product added successfully"  
}
```

Sales Management APIs

POST /api/sales/add

Purpose: Record a new sales transaction.

Input:

```
{  
  "product_id": 1,  
  "quantity": 5,  
  "amount": 500  
}
```

Expected Output:

```
{  
  "status": "success",  
  "message": "Sale recorded successfully"  
}
```

GET /api/sales/report

Purpose: Retrieve sales reports and statistics.

Input: None

Expected Output:

```
{  
  "total_sales": 25000,  
  "transactions": 120  
}
```

Expense Management APIs

POST /api/expenses/add

Purpose: Record business expenses.

Input:

```
{  
  "category": "Marketing",  
  "amount": 5000  
}
```

Expected Output:

```
{  
  "status": "success",  
  "message": "Expense added successfully"  
}
```

GET /api/expenses/report

Purpose: Retrieve expense reports.

Input: None

Expected Output:

```
{  
  "total_expense": 15000  
}
```

AI APIs

POST /api/ai/analytics

Purpose: Analyze business performance using AI.

Input:

```
{  
  "sales_data": "...",  
  "expense_data": "..."  
}
```

Expected Output:

```
{  
  "insight": "Sales increased by 15% this month."  
}
```

POST /api/ai/recommendation

Purpose: Generate AI-based business recommendations.

Input:

```
{  
  "business_data": "..."  
}
```

Expected Output:

```
{  
  "recommendation": "Increase inventory for Product A due to high demand."  
}
```

GET /api/dashboard/summary

Purpose: Display key business metrics on the dashboard.

Input: None

Expected Output:

```
{  
  "total_sales": 50000,  
  "total_expenses": 20000,  
  "profit": 30000  
}
```

AI Integration Workflow

Why AI is Needed

Business owners often collect large amounts of sales, inventory, and expense data but may struggle to identify trends and make informed decisions. Artificial Intelligence helps analyze business data automatically and provides actionable insights and recommendations, reducing manual analysis and improving decision-making.

AI Service / Model

- OpenAI API (GPT Model) or Google Gemini API
- Integrated through the WordPress plugin backend using API requests

Input to the AI System

The AI system receives:

- Sales records
- Expense data
- Inventory levels
- Product performance statistics
- Business activity reports

Example Input

- Monthly sales: \$10,000
- Monthly expenses: \$7,000
- Product A stock: 15 units remaining
- Product A sales growth: 25%

Output from the AI System

The AI generates:

- Business performance summaries
- Inventory restocking recommendations
- Sales trend analysis
- Expense reduction suggestions
- Business growth insights

Example Output

"Product A sales have increased by 25% this month. Current inventory is low and restocking is recommended within the next week. Marketing expenses have increased by 15% compared to the previous month."

AI Workflow

Business Data

↓

WordPress Plugin Backend

↓

OpenAI API / Gemini API

↓

AI Analysis & Recommendation Generation

↓

Business Insights Report

↓

Admin Dashboard Display

How AI Improves the Project

- Automates business data analysis
- Reduces manual reporting effort
- Helps identify sales and inventory trends
- Supports data-driven decision-making
- Provides intelligent recommendations for business growth
- Improves overall operational efficiency

References

1. WordPress Developer Documentation. Available: <https://developer.wordpress.org>
2. OpenAI API Documentation. Available: <https://platform.openai.com/docs>
3. Google AI. Gemini API Documentation. Available: <https://ai.google.dev/docs>
4. Pressman, R. S., & Maxim, B. R. *Software Engineering: A Practitioner's Approach*, 9th Edition. McGraw-Hill Education, 2019.